

Valid parentheses.

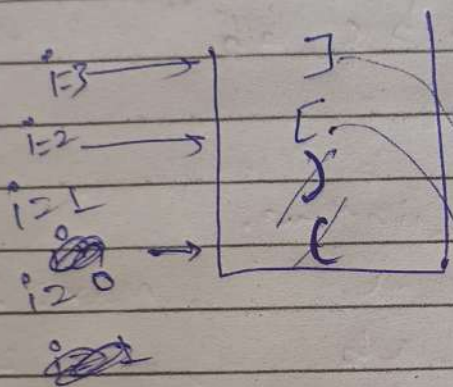
Q.15

$S = "() [] \{ \}"$

$\uparrow \uparrow$
 $\uparrow \uparrow$

Using stack data structure. (LIFO)

ek loop chalega ($i=0, i < s.length, i++$)



opening hai aur ek
closing hai to pop
karenge aur
stack empty ho
jayega.

phir dono pop ho
jayega aur stack
empty ho jayega.

→ Agar last element ek baar stack
empty hua to return true karenge.

* Base cases:-

→ agar string pairs me nhi hai to return
false, or even number nhi hai to false.

→ $ch = ') '$ & top ' (', return false.

ex:- $S = "([[]])"$ → return false.

facing opposite
to each other

// Code .

class Soln {

public boolean isValid (String s) {

int n = s.length();

if (n % 2 != 0) {

return false;

}

Stack<Character> ~~char~~^{stack} = new Stack<>();

for (int i = 0; i < n; i++) {

char ch = s.charAt(i);

if (ch == '(' || ch == '[' || ch == '{') {

stack.push(ch);

} else {

if (stack.empty()) {

return false;

}

char top = stack.peek();

if (ch == ')' && top != '(') {

return false;

} else if (ch == ']' && top != '[') {

return false;

} else if (ch == '}' && top != '{') {

return false;

} else {

stack.pop();

}

}

}

return (stack.size() == 0);

}

}